

# CHEAT, NLTK, etc:

## text analytics in Python

Introduction to Python tools for text analytics, and  
the CHEAT approach to a text analytics challenge  
MorphoChallenge

- Website: <https://www.nltk.org/>
- Texts:
  - E Atwell. 2005. Combinatory Hybrid Elementary Analysis of Text [atwell05cheat.pdf](#)
  - S Bird. 2006. NLTK: The Natural Language Toolkit [bird06nltk.pdf](#)

# Text Analytics in Python

- SketchEngine, Weka etc enable linguists to try text analytics experiments without programming
- To develop and test a new ML algorithm, you need to write code, eg in Python using NLTK
- Example from 2005: CHEAT.py
- Don't code from scratch: Google code archive, NLTK, etc ... Smart program can be a few lines
- Don't code unless you have to: SketchEngine etc

# NLTK Natural Language Toolkit

<https://www.nltk.org/>

”NLTK is a leading platform for building Python programs to work with human language data. It provides easy-to-use interfaces to [over 50 corpora and lexical resources](#) such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries, and an active [discussion forum](#).”



# NLTK Natural Language Toolkit

<https://www.nltk.org/>

“... hands-on guide introducing programming fundamentals alongside topics in computational linguistics, plus comprehensive API documentation”

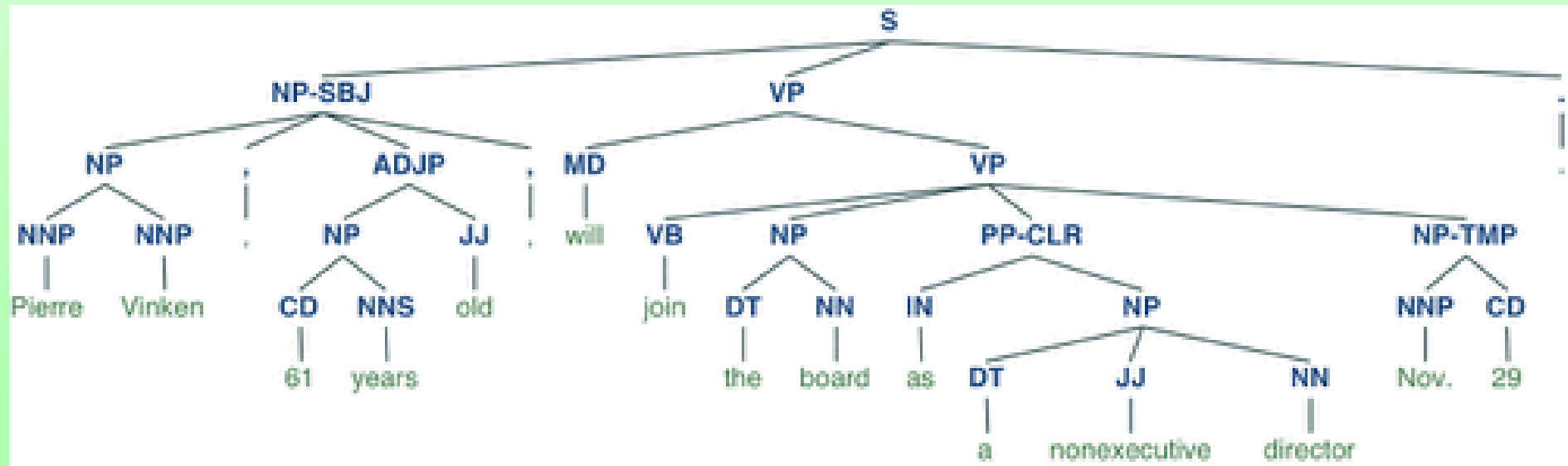
... [Natural Language Processing with Python](#) book

... tools for testing and display ...



# NLTK: display a parse tree

```
>>> from nltk.corpus import treebank
>>> t = treebank.parsed_sents('wsj_0001.mrg')[0]
>>> t.draw()
```



# Other Python NLP tools

SpaCy <https://spacy.io/>

Specific NLP tasks: [Gensim](#) for Topic Modelling

Surveys of Python NLP toolkits

<https://kleiber.me/blog/2020/08/26/top-10-python-nlp-libraries-2020/>

<https://towardsdatascience.com/python-libraries-for-natural-language-processing-be0e5a35dd64>



# shared tasks for AI research

AI / ML / CL conferences invite researchers to present their research

May include “shared tasks” to promote research  
eg SemEval, MorphoChallenge

<http://morpho.aalto.fi/events/morphochallenge/>

Unsupervised ML segmentation of words into morphemes

 New ML algorithms, not available in toolkits;  
coded in programming language, eg Python

# MorphoChallenge2005, CHEAT

First MorphoChallenge in 2005: MSc coursework

Some Leeds students entered the real contest

CHEAT ensemble combined student results

Youtube started 2005, max 10 minute “TikToks”

<https://www.youtube.com/watch?v=Uoj6RjZpAkk>

Academics set up VideoLectures.net for lectures

[http://videlectures.net/pcw06\\_atwell\\_cheat/](http://videlectures.net/pcw06_atwell_cheat/)



- CHEAT 2005 ppt, sorry for crackly sound!

# **Combinatory Hybrid Elementary Analysis of Text: the CHEAT approach to MorphoChallenge2005**

**Eric Atwell  
School of Computing  
University of Leeds  
Leeds LS2 9JT**

**Andrew Roberts  
Pearson Longman  
Edinburgh Gate  
Harlow CM20 2JE**



# With the help of Eric Atwell's Computational Modelling MSc class...

- Khurram AHMAD
- Rodolfo ALLENDES OSORIO
- Lois BONNIER
- Saad CHOUDRI
- Minh DANG
- Gerard David HOWARD
- Simon HUGHES
- Iftikhar HUSSAIN
- Lee KITCHING
- Nicolas MALLESON
- Edward MANLEY
- Khalid Ur REHMAN
- Ross WILLIAMSON
- Hongtao ZHAO



# **Our guiding principle: get others to do the work**

**PLAGIARISM is BAD ... but  
in Software Engineering, REUSE is GOOD !  
We can't just copy results from another entrant ... but  
we may get away with smart copying**

**We can copy results from MANY systems, then use  
these to “vote” on analysis of each word**

**BUT – how can we get results from other contestants?  
... set MorphoChallenge as MSc coursework, students  
must submit their results to lecturer for assessment!**



# **But is this really “unsupervised learning”?**

**“... the program cannot be given a training file containing example answers...”**

**Our program is given several “candidate answer files”, BUT does not know which (if any) is correct**

**So it IS unsupervised learning; moreover, it is...**



# **Triple-layer Super-Sized Unsupervised Learning:**

- Unsupervised Learning by students**
- Unsupervised Learning by student programs**
- Unsupervised Learning by `cheat.py`**



# **Unsupervised Learning by students**

- **Eric Atwell gave background lectures on Machine Learning, and Morphological Analysis**
- **Students were NOT give “example answers”: unsupervised morphology learning algorithms**
- **So, student learning was Unsupervised Learning**



# **Unsupervised Learning by student programs**

- **Pairs of students developed MorphoChallenge entries, e.g.:**
  - **Saad CHOUDRI and Minh DANG**
  - **Khalid REHMAN and Iftikar HUSSAIN**
- **Student programs were “black boxes” – we just needed results**



# **Unsupervised learning by cheat.py**

- **Read outputs of other systems,  
line by line**
- **Select majority-vote analysis**
- **If there is a tie, select result from  
best system (highest F-measure)**
- **Output this – “our” result!**



# **cheat.py and cheat2.py**

**This worked in theory, but...**

**... some student programs re-ordered  
the wordlist, so outputs were not  
aligned, like-with-like**

**Andrew Roberts developed more robust  
cheat2.py, which REALLY worked!**



# **Results: cheating works!**

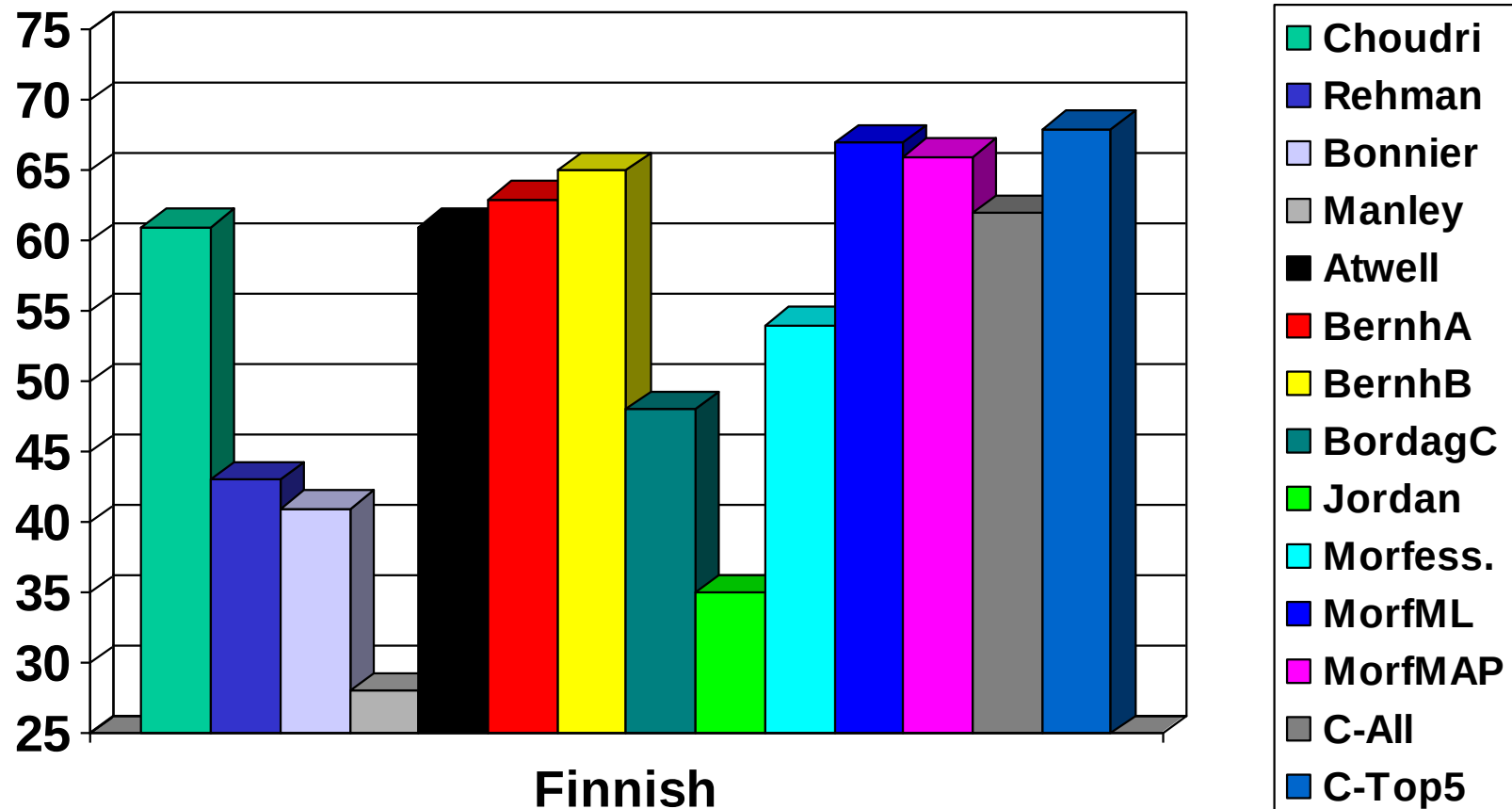
**See results tables in the full paper.**

**For all 3 languages (English, Finnish, Turkish), our cheat system scored a higher F-measure than any of the contributing systems!**

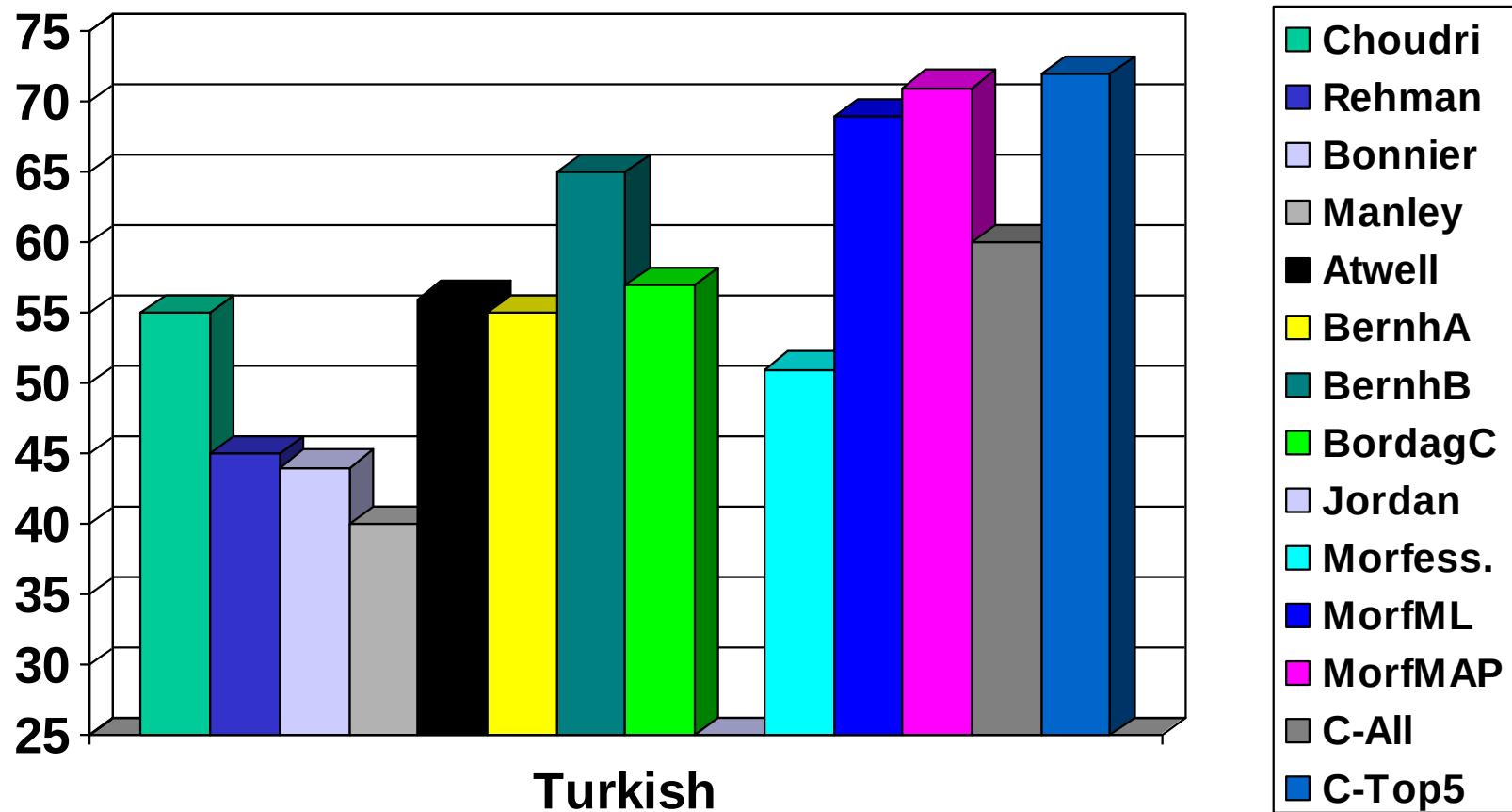
**?? We added Morfessor output, this did not change our scores !! Maybe there is something fishy going on?**



# F-measure with reference algorithms



# F-measure with reference algorithms



# **Conclusions: Machine Learning and Student Learning**

**cheat.py is actually a committee of unsupervised learners, used previously in ML (Banko and Brill 2001)**

**(but we didn't learn this from the literature till afterwards – a fourth layer in Super-Sized Unsupervised Learning?)**

**BUT cheat is also a novel idea in Student Learning: get students to implement the learners, so students learn (about ML as well as domain: in this case, morphology)**

**MorphoChallenge inspired our students to produce outstanding coursework!**



# **Thank you!**

**We'd like to thank the MorphoChallenge  
organisers for an inspiring contest!**

**And thanks to the audience for sitting  
through our presentation**

**Eric Atwell [eric@comp.leeds.ac.uk](mailto:eric@comp.leeds.ac.uk)**

**Andrew Roberts**

**[andrew.roberts@pearson.com](mailto:andrew.roberts@pearson.com)**

